

Unit 5 :: Fundamentals of Data Visualization

☞ SEC-03-PHY-91-R-03 ☞

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Syllabus

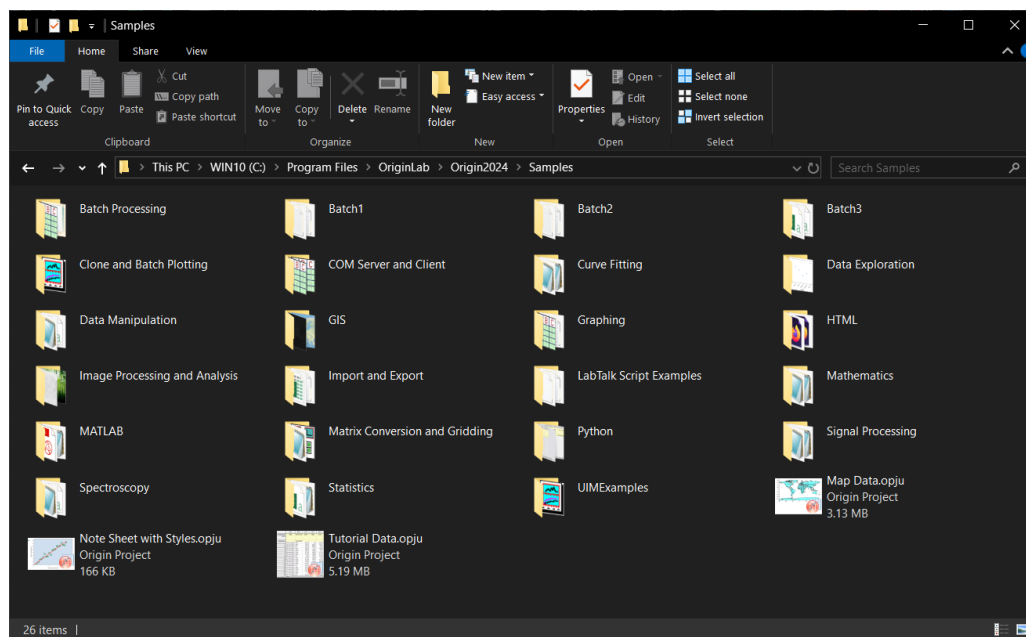
Unit 5: Fundamentals of data visualization

Principles and use cases for categorical and continuous data plots, including column, line, pie, bar, area and scatter plots; customizing chart elements (axes, scales, legends, and labels); fine-tuning aesthetics for clarity and technical accuracy; advanced plotting techniques: 3D surface plots, contour mapping and multi-layered/multipanel figures.

1 Fundamentals of plotting in Origin

An installation of Origin comes with a collection of sample data which is recommended for a beginner to learn the fundamentals of plotting and subsequent customization. In order to access this collection of sample data, one can go to the main menu and click Help > Open Folder > Sample Folder.

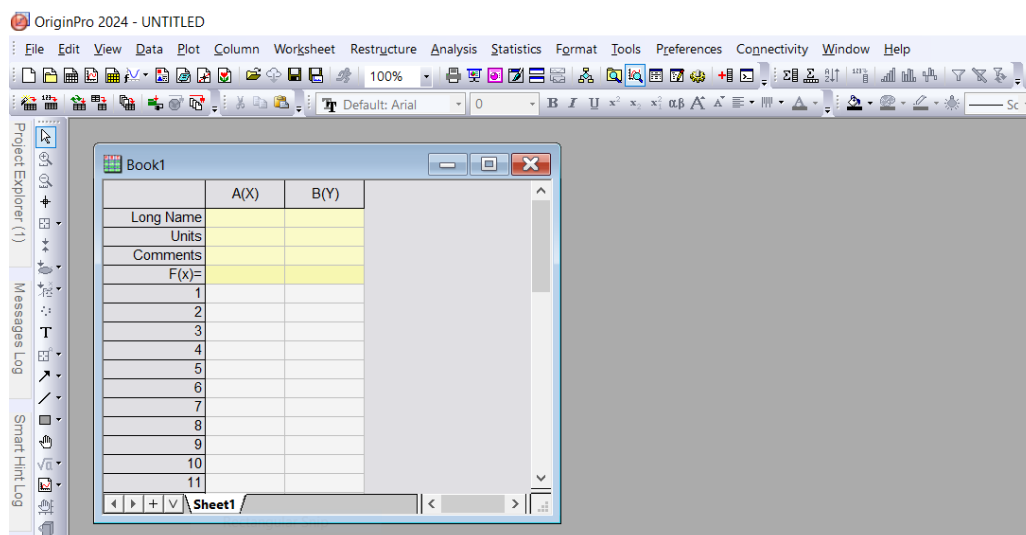
This would open up the Sample folder in a Windows Explorer.



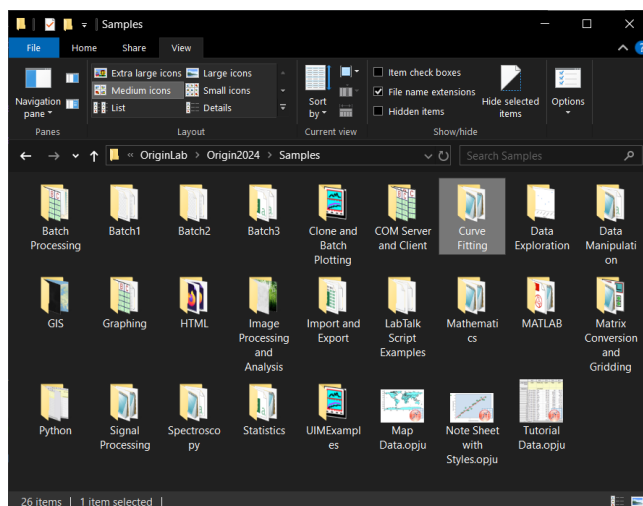
1.1 Plotting the first graph

1.1.1 Importing data from sample folder

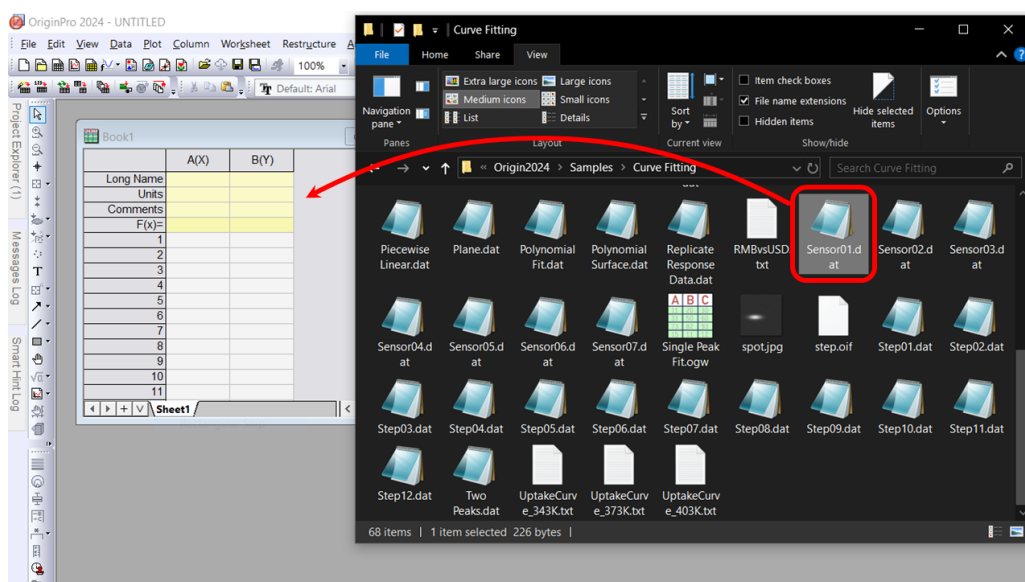
- I. Launch Origin to find the default empty workbook which contains a single worksheet with two columns.



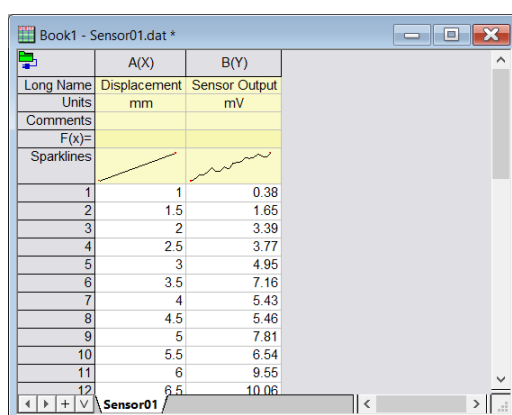
II. Navigate into the Curve Fitting folder in the Sample folder.



III. Locate any of the DAT files, such as Sensor01.dat. Then drag and drop it into the Origin workbook.

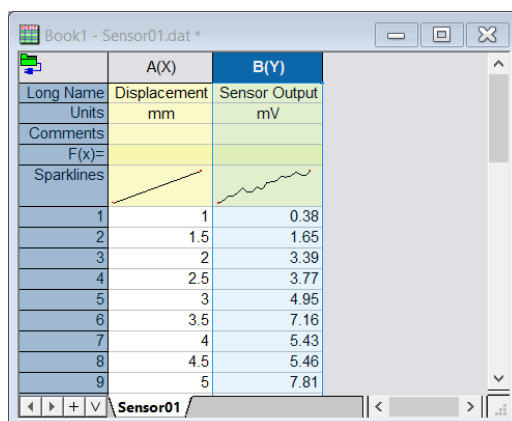


IV. This would import the data from the DAT file into the workbook.



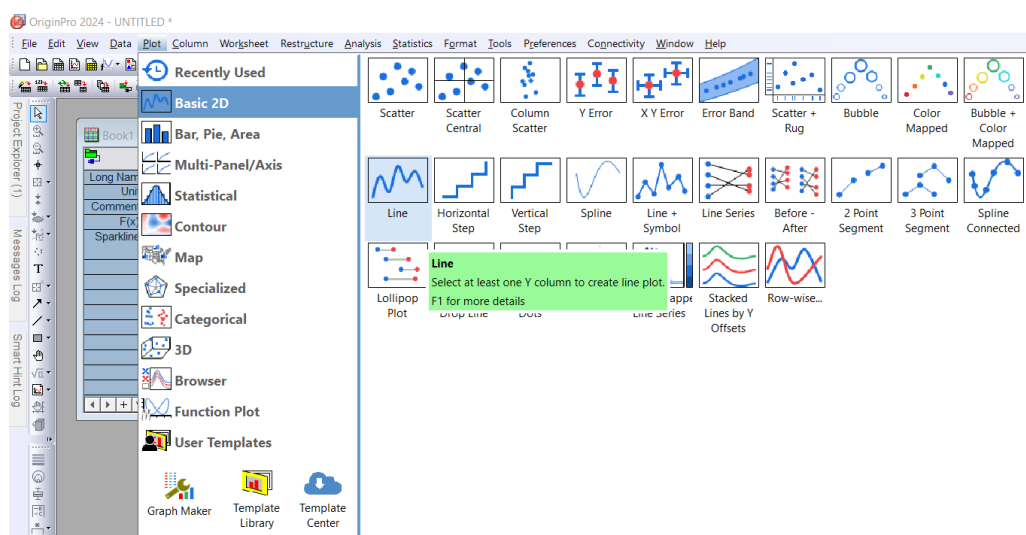
1.1.2 Plotting the graph

I. Click on the header of column B to select the entire column.

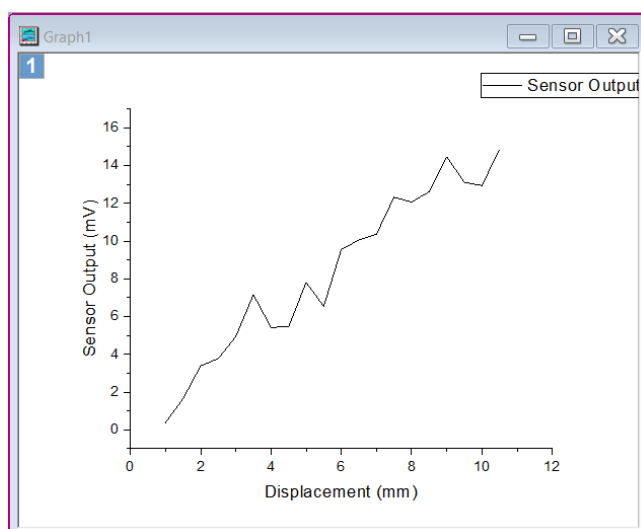


	A(X)	B(Y)
Long Name	Displacement	Sensor Output
Units	mm	mV
Comments		
F(x)=		
Sparklines		
1	1	0.38
2	1.5	1.65
3	2	3.39
4	2.5	3.77
5	3	4.95
6	3.5	7.16
7	4	5.43
8	4.5	5.46
9	5	7.81

II. From the main menu, we click Plot > Basic 2D > Line.

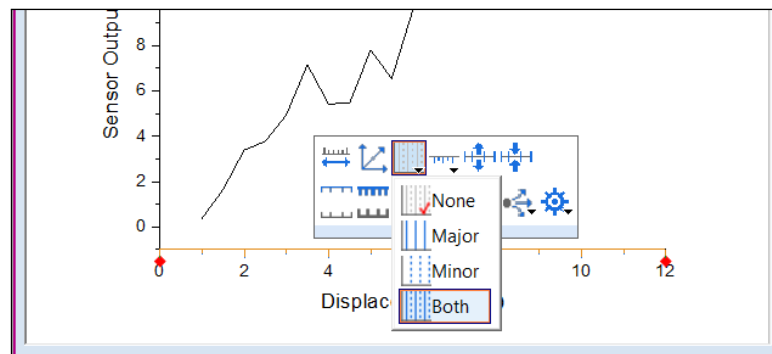


III. This produces a basic 2D plot.

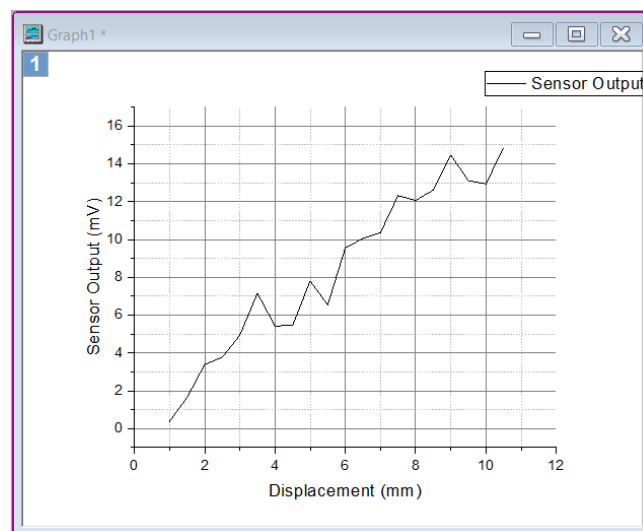


1.1.3 Basic customization of the plot

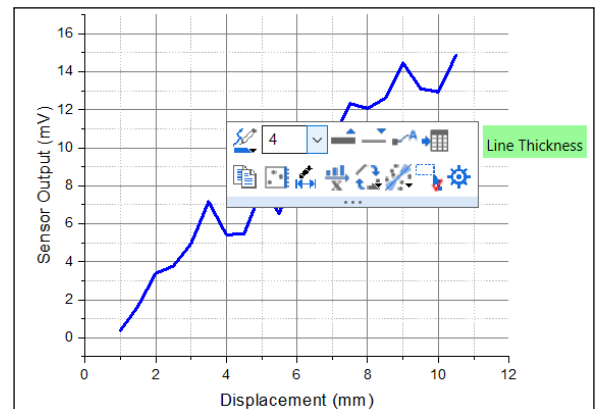
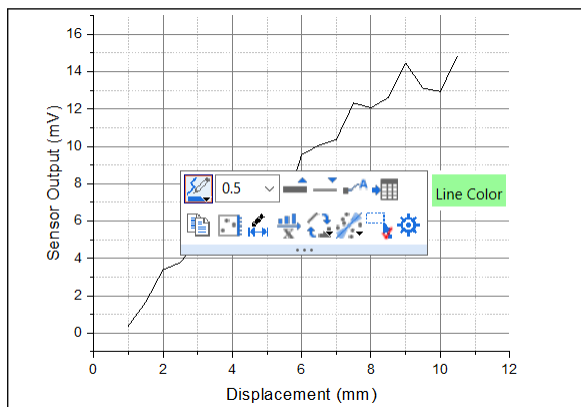
I. Click on the x -axis so that the mini toolbar pops up. Then click Show Gridlines > Both.



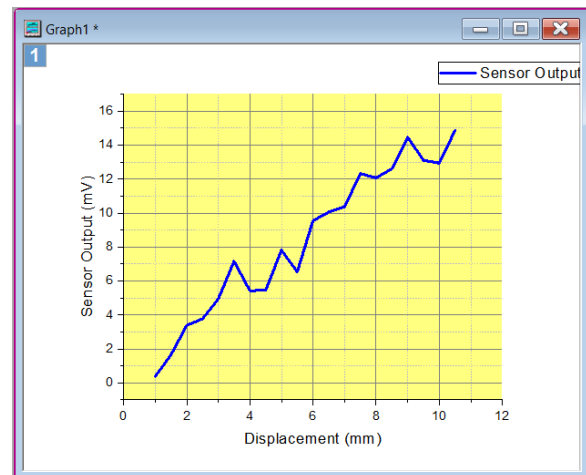
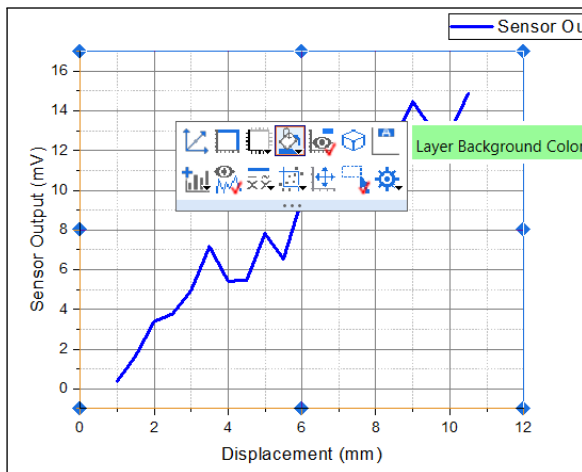
II. Repeat the above step on the y -axis to get a plot with major and minor gridlines.



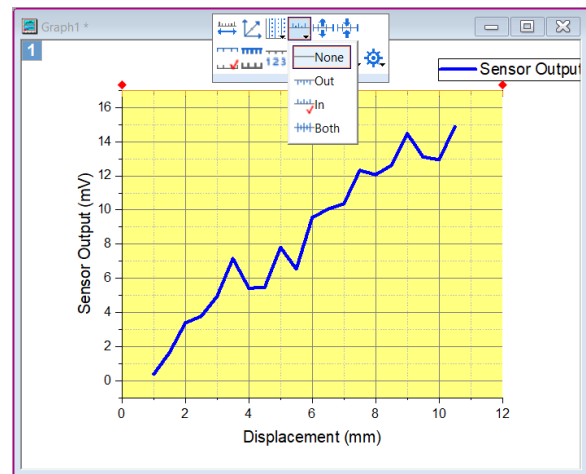
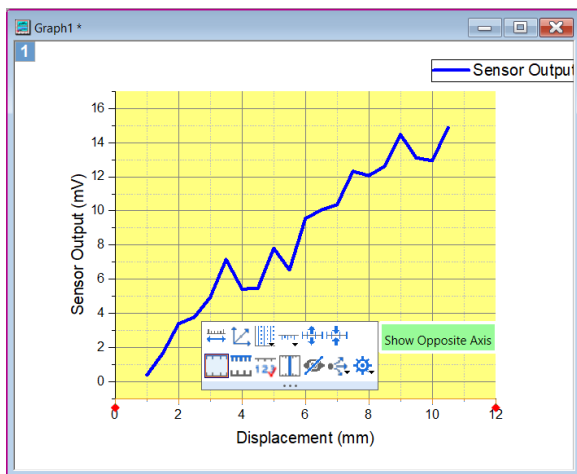
III. Click on the line plot so that the mini toolbar pops up. Then click on the Line Color button to select a colour, say blue. Also set the Line Thickness to a larger value, say 4.



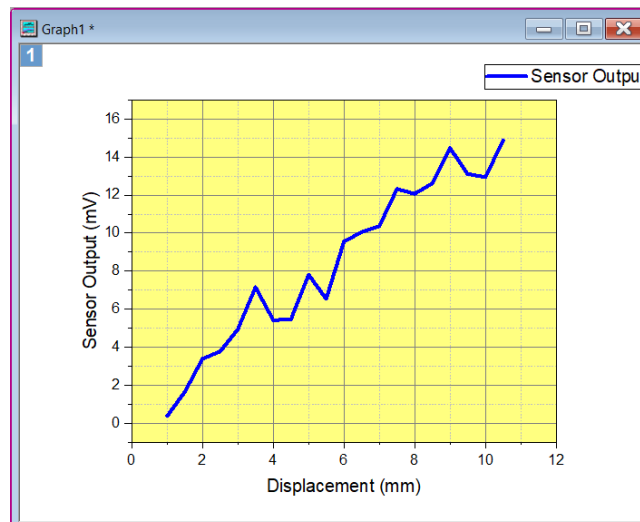
IV. Click on the plot background (ensure that the plot or the gridlines are not clicked), click on Layer Background Color and then select some background colour, such as light yellow.



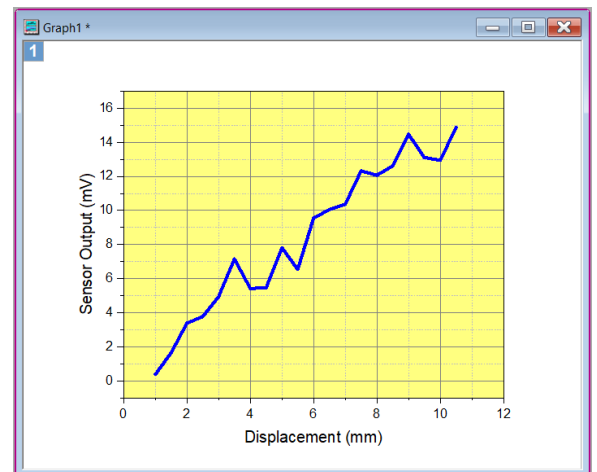
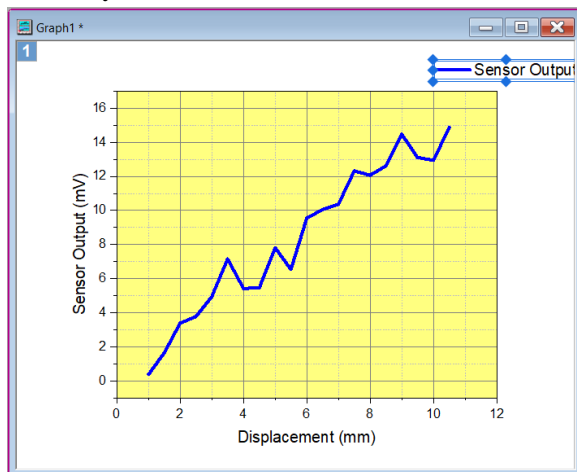
V. We can create a bounding box by displaying opposite axes and removing the tick marks. To do so, we click on any of the tick labels on the x -axis and then click Show Opposite Axis on the pop-up menu. On the pop-up menu of the opposite axis, we click on the Tick Style button and then on None.



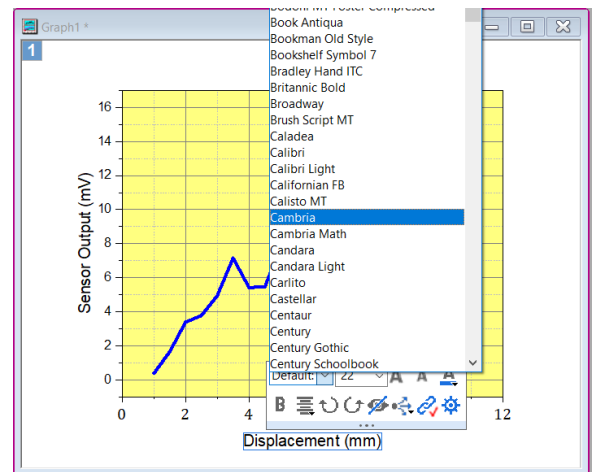
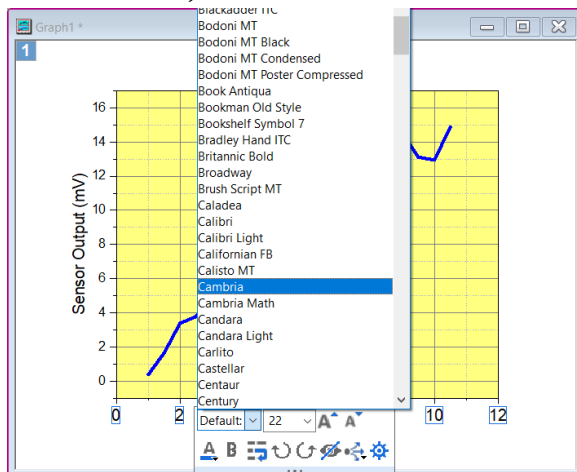
VI. Repeating the above step for the y -axis gives us the desired bounding box around the graph.



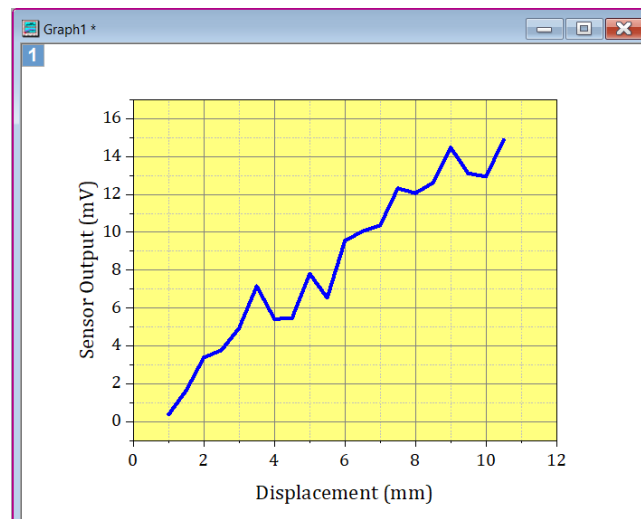
VII. We can remove the graph legend by clicking on it and pressing the delete button on the keyboard.



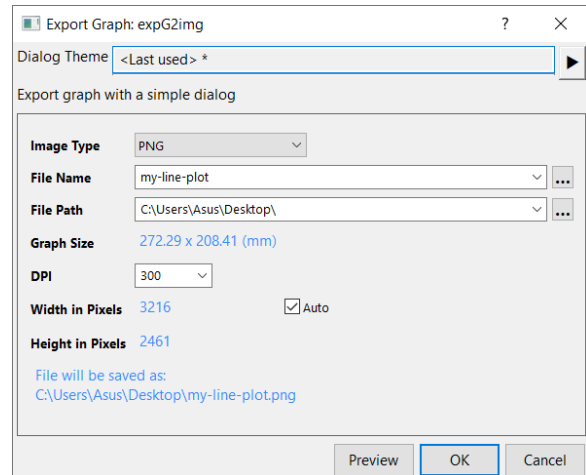
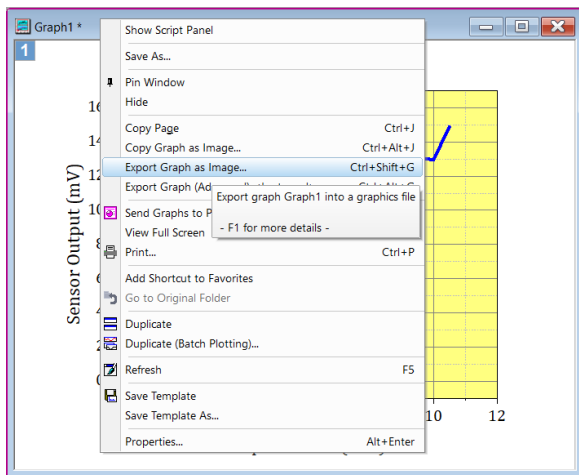
VIII. To modify the typeface of the tick labels, we click on any of the tick labels and on the pop-up menu, we set the font size (here, 22) and the font (here, Cambria). Likewise, we click on the axis label and on the pop-up menu, we set the font size (here, 26) and the font (here, Cambria).



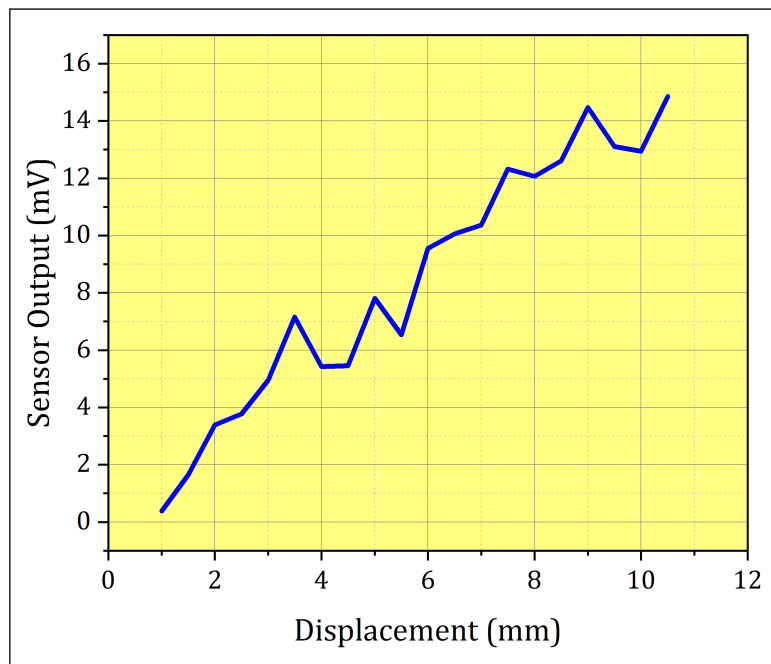
IX. We repeat the above step for the y-axis tick and axis labels to obtain a graph with the desired aesthetics.



- X. Finally, export the graph into an image file, we right-click on the graph's title bar, and then click on **Export Graph as Image**. Then in the export graph dialog, we set **Image Type** = PNG, **File Name** as appropriate (here, my-line-plot), and **File Path** to be the appropriate folder (here, Desktop). Clicking on **OK** will export the graph into a PNG file.



- XI. Finally we obtain the plot file which can now be included in a document as required.

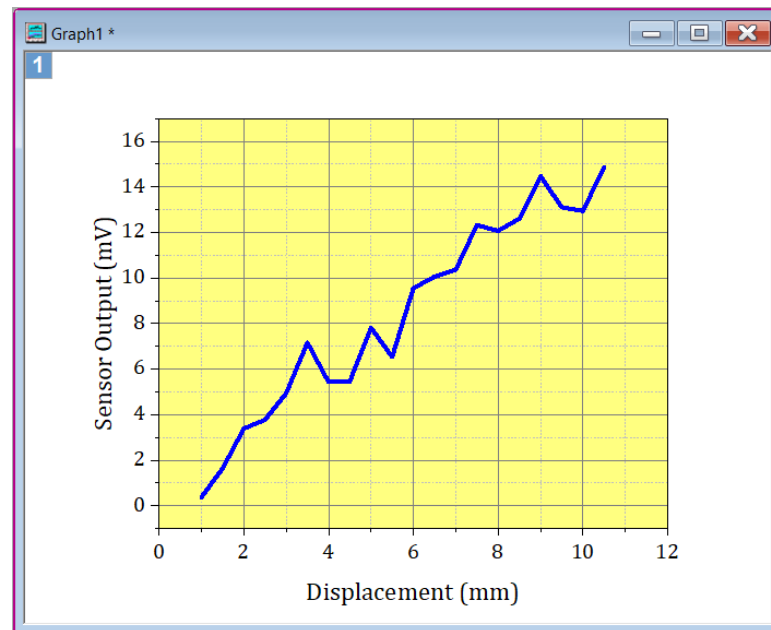


1.2 Using graph templates

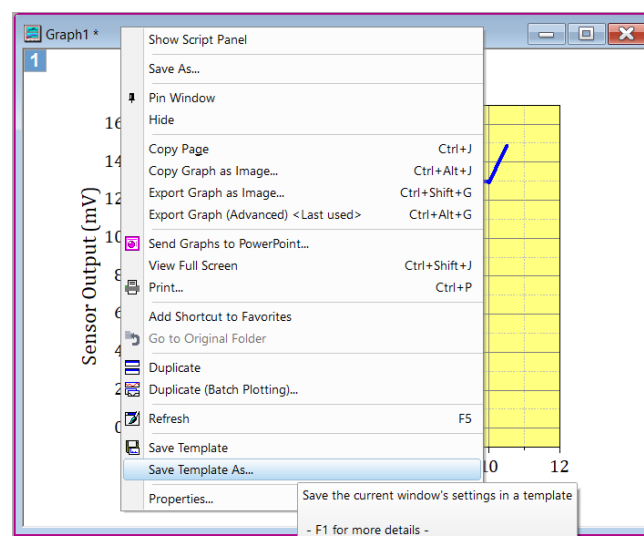
A template is a pre-made pattern, model, or mold used as a guide to make multiple identical or similar things. Origin allows us to prepare a graph template based on a graph which is customized according to our requirements. Then later, we can use this template to quickly make identical customizations on other graphs.

1.2.1 Creating a graph template

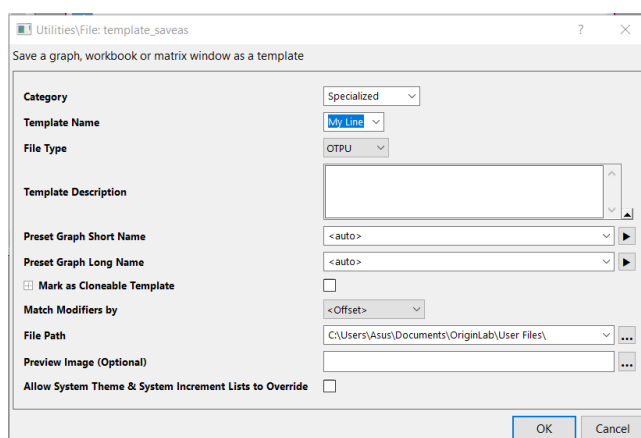
I. We start with a graph that we have customized as per our requirement.



II. To use this graph as a template, we right-click on the graph's title bar, and then click on Save Template As... from the context menu.

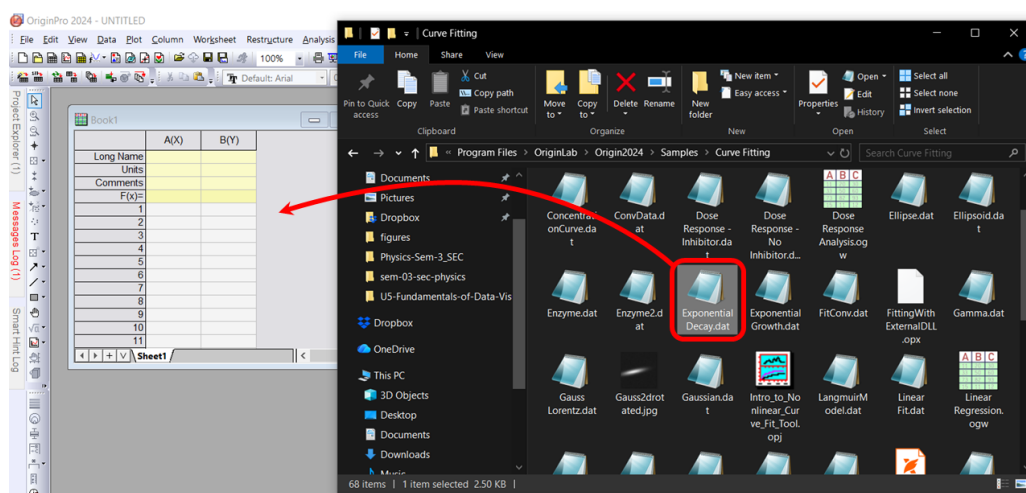


- III. In the save template dialog, we enter the name of the template, for example Template Name = My Line. Then we click OK.

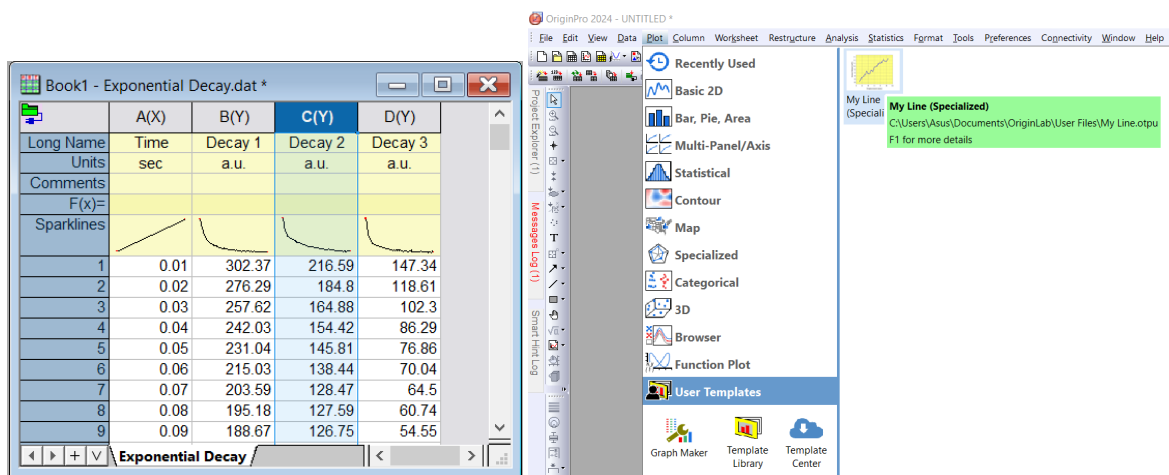


1.2.2 Applying a graph template

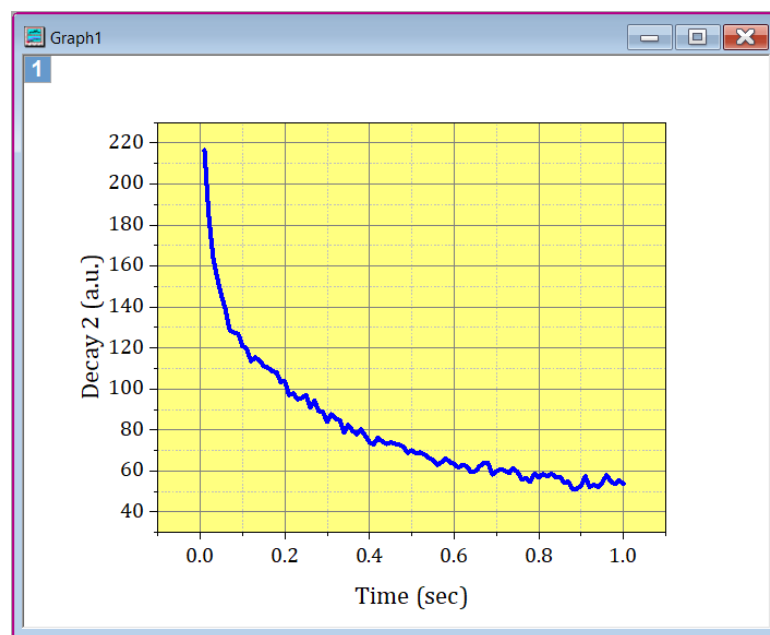
- Suppose in a different Origin workbook, we wish to plot some other graph using the above template. For example, we can drag and drop the Exponential Decay .dat file from the Samples > Curve Fitting folder into the workbook.



2. In order to plot the data in column C, we click on that column's header. Then from the main menu, we click Plot > User Templates > My Line.



3. This gives us the desired plot which is customized as per the saved graph template. We can find that the plotting style of this graph is identical to the one which we started with.



2 Categorical and continuous data plots

A dataset is said to be *categorical* if its independent variable takes one of a small, discrete set of labels (e.g. the day of the week, the floor number of a skyscraper), and *continuous* if its independent variable can take any value within a range (e.g. time, angle, or magnetic field strength). The choice of plot type should follow from which of these two kinds of variable is being plotted, not from visual preference alone.

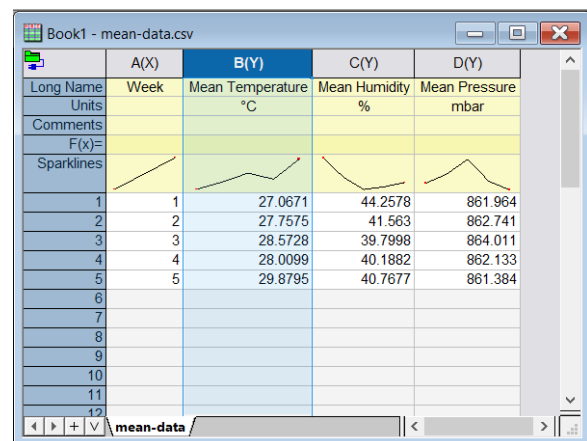
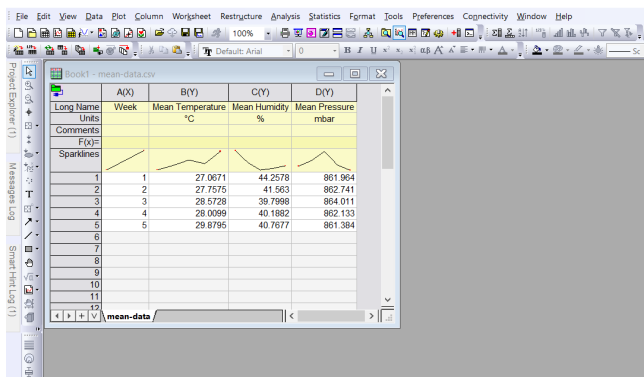
2.1 Column plots

In a *column plot*, each category is represented as a vertical bar whose height is proportional to a summary value (such as a mean, a total or a count).

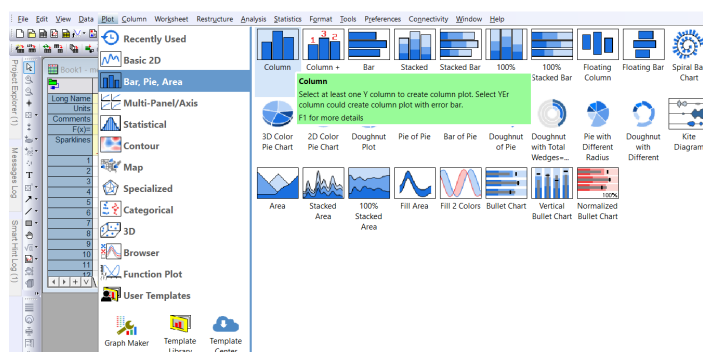
2.1.1 Creating a column plot in Origin

As a demonstration, we will create a column plot using the file named `mean-data.csv`, which has recorded the average temperature, humidity and pressure data for five weeks.

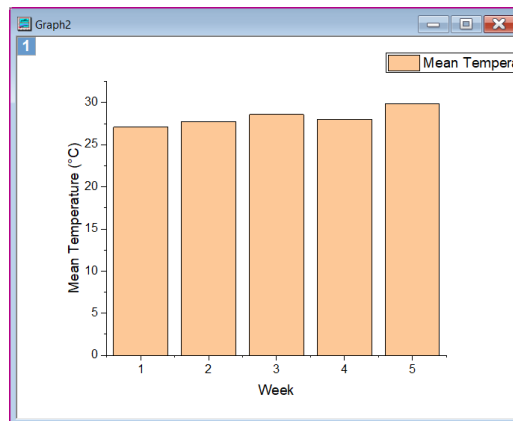
- I. We open the CSV file as described in the previous unit. To create the column plot for the mean temperatures for five weeks, first we select the entire column by clicking on the column name.



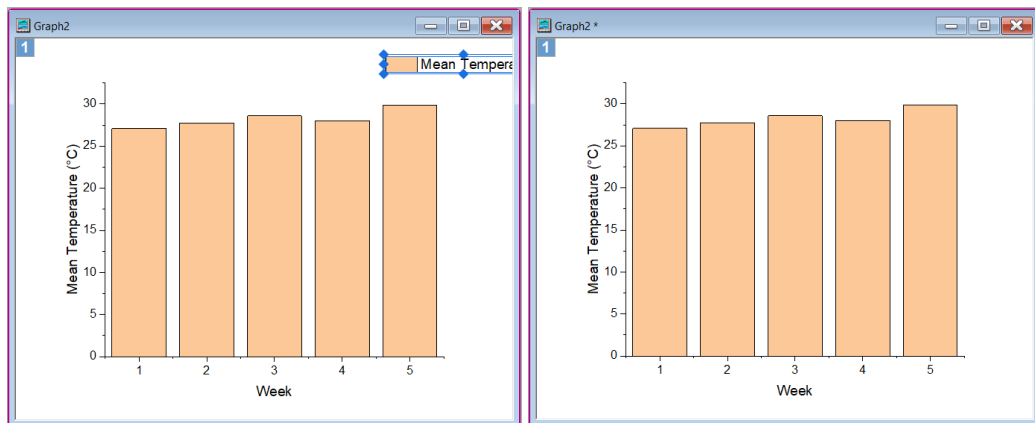
- II. From the main menu, we click Plot > Bar, Pie, Area, and then click Column.



III. This produces a rudimentary column plot with default settings.



IV. Because this is a column plot of only the mean temperature, we can click on the legend at the top right and press the delete button to remove it.



V. Because the column bars are about similar heights, we are unable to notice significant differences between them if we plot them from zero at the bottom. So we need to change the scale of the y-axis values in a manner such that their differences become more pronounced.

To do so, we click on any of the y-tick labels and then click the `Open Properties Dialog` logo.



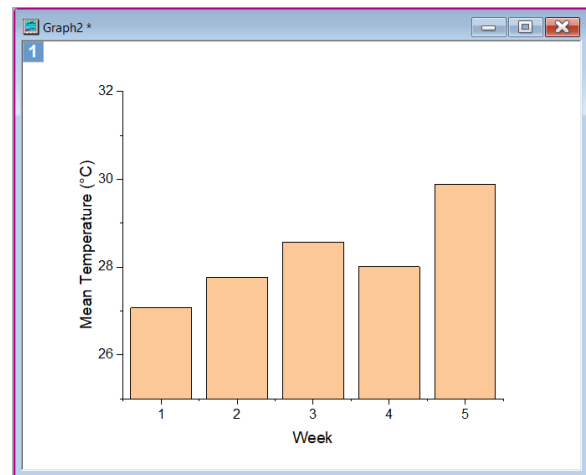
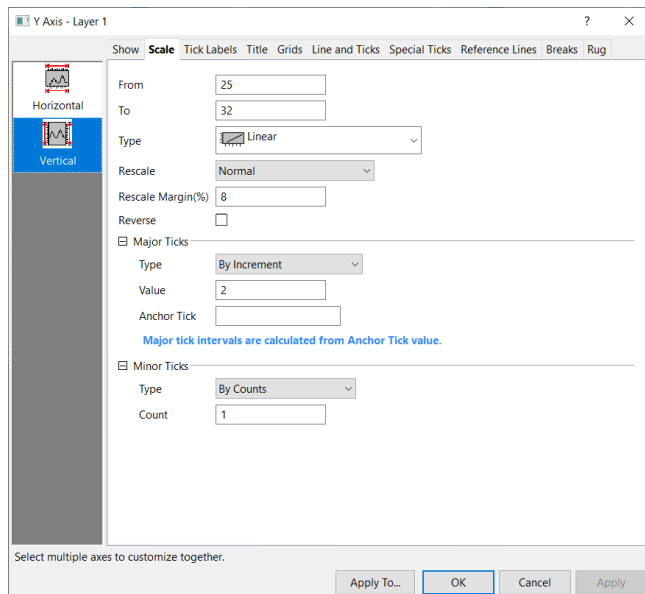
VI. In the Properties dialog, we click on the Scale tab and approximately set the following values as From = 25, To = 32, so as to ensure that the y -axis spans the values from 25 to 32.

We also set the major ticks as

Type = By Increment
Value = 2

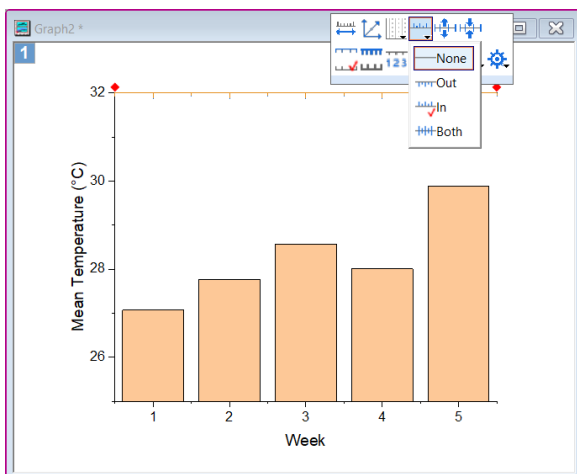
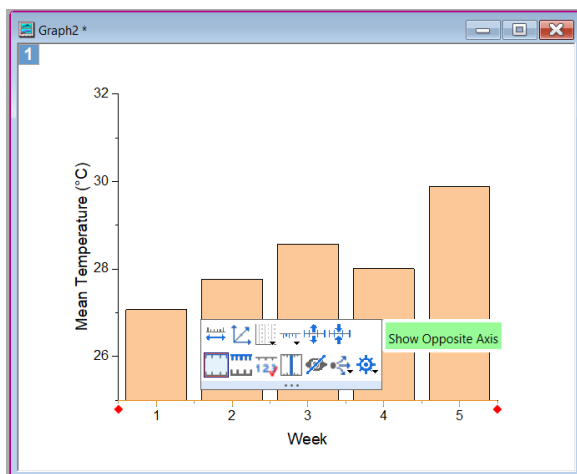
and the minor ticks as

Type = By Counts
Value = 1

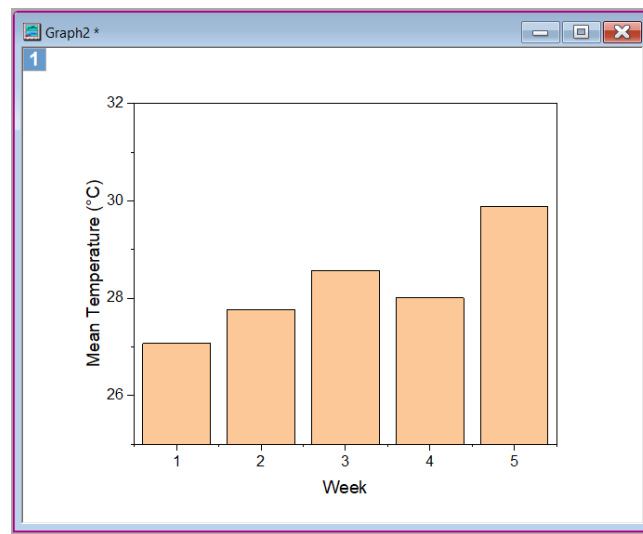


This shows much more discernible differences between the mean temperatures.

VII. Finally to place the graph within a rectangular bounding box, first we need to enable opposite axes by clicking on any of the axes lines, say the x -axis line, and then clicking on the Show Opposite Axis button, so that an identical axis appears at the top. Then we click on the new axis at the top, then click on Axis Type button and lastly click on None.



VIII. We repeat the same process for the y -axis as well to finally obtain the required column plot within a bounding box.



2.2 Line plots

A *line plot* connects successive data points with straight (or smoothed) line segments. It is the standard choice for a continuous independent variable, and not for categorical data.

2.2.1 Creating a line plot in Origin

The creation of a line plot is described in detail in sub-section 1.1.

2.3 Pie charts

A *pie chart* represents each category's share of a whole as a wedge of a circle, with wedge angle proportional to that category's proportion of the total.

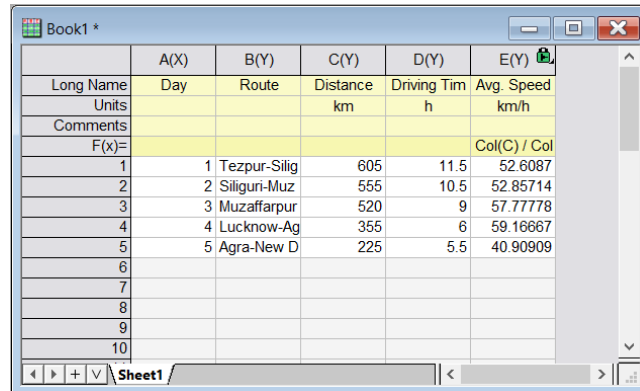
2.3.1 Creating a pie chart in Origin

As a demonstration, let us consider a car being driven from Tezpur to New Delhi over a span of five days. The details of the travel are as per the table given below:

Day	Route	Distance (km)	Driving time (h)	Avg. speed (km/h)
1	Tezpur–Siliguri	605	11.5	52.6
2	Siliguri–Muzaffarpur	555	10.5	52.9
3	Muzaffarpur–Lucknow	520	9.0	57.8
4	Lucknow–Agra	355	6.0	59.2
5	Agra–New Delhi	225	5.5	40.9

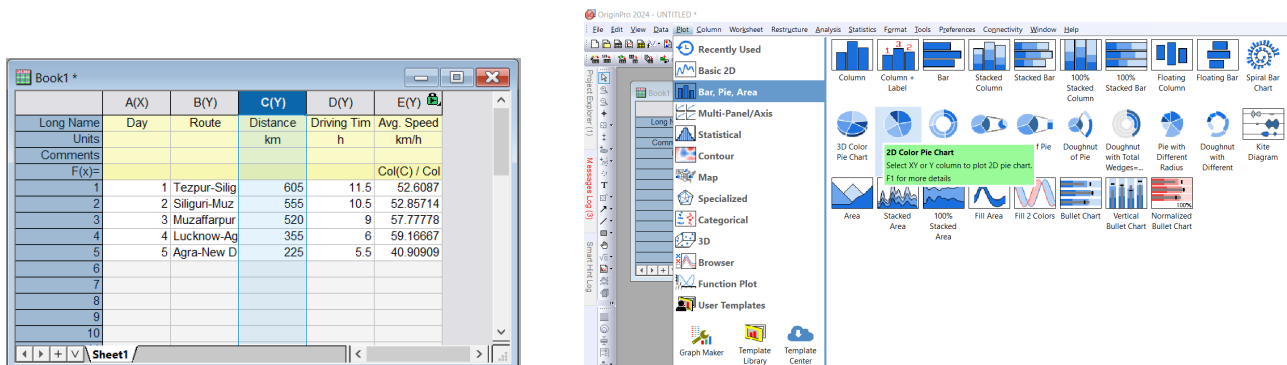
Because the distance and driving time data for each day represent shares of the whole of total distance travelled and total time driven respectively, we can represent both of these using pie charts.

I. We enter the data given in the table above into a workbook.

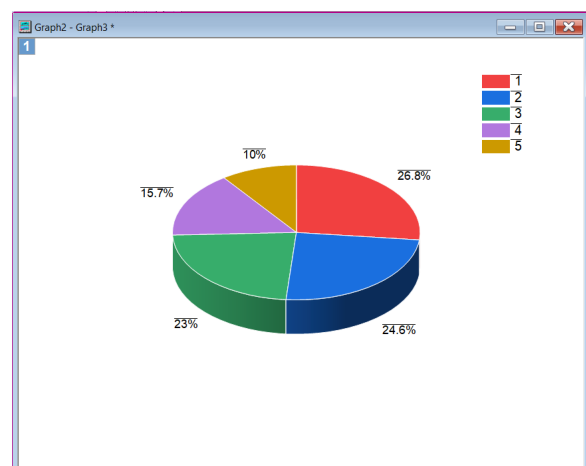
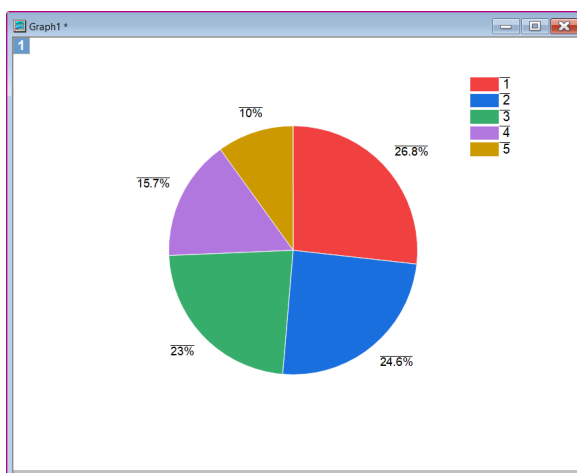


	A(X)	B(Y)	C(Y)	D(Y)	E(Y)
Long Name	Day	Route	Distance	Driving Tim	Avg. Speed
Units			km	h	km/h
Comments					
F(x)=					Col(C) / Col
1	1	Tezpur-Silig	605	11.5	52.6087
2	2	Siliguri-Muz	555	10.5	52.85714
3	3	Muzaffarpur	520	9	57.77778
4	4	Lucknow-Ag	355	6	59.16667
5	5	Agra-New D	225	5.5	40.90909
6					
7					
8					
9					
10					

II. To create a pie chart for the day-wise distance data, we click on the header of column C to select the entire column. Then from the main menu, we click Plot > Bar, Pie, Area > 2D Color Pie Chart.



III. This gives us the desired 2D pie chart. If we choose 3D Color Pie Chart instead, we would obtain a pie chart with 3D effect instead.

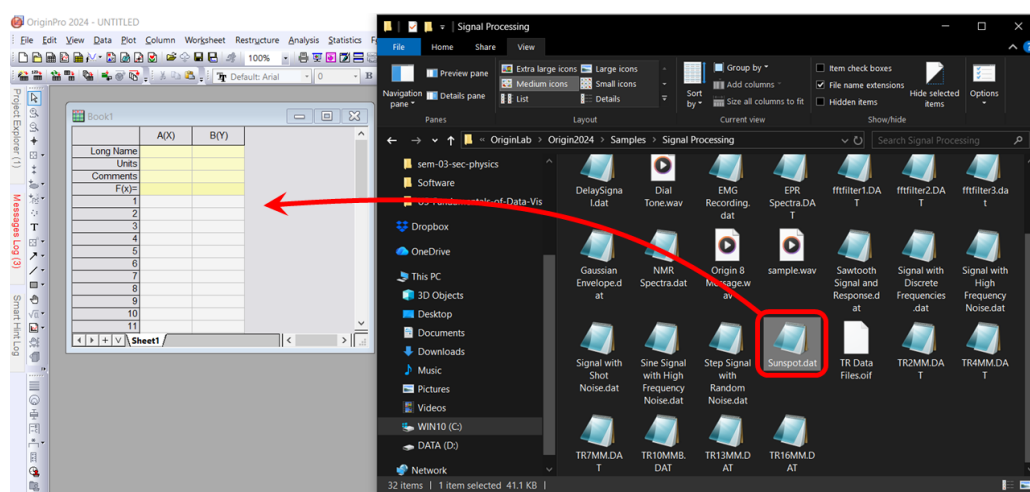


2.4 Scatter plots

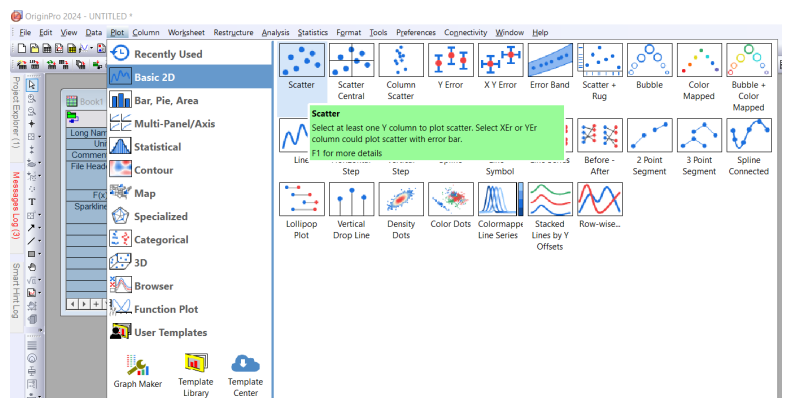
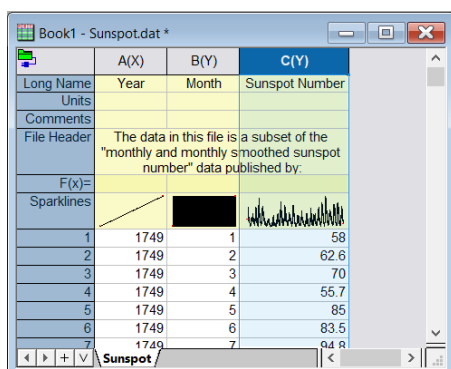
A *scatter plot* represents each observation as an individual point, with no line joining successive points. It is the correct choice when the relationship between two continuous variables is of interest, rather than the trend of one variable against time. A scatter plot is particularly used when there is no reason to assume the points are ordered or continuous in the underlying physical process.

2.4.1 Creating a scatter plot in Origin

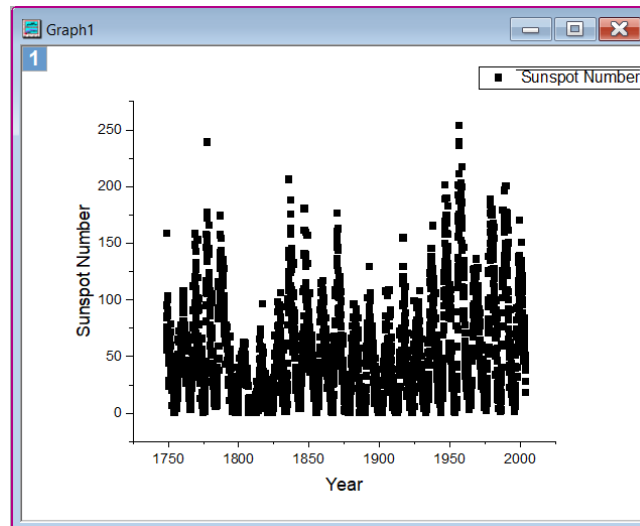
- I. Suppose in a different Origin workbook, we wish to create a scatter plot using some other data. For example, we can drag and drop the Sunspot .dat file from the Samples > Signal Processing folder into the workbook.



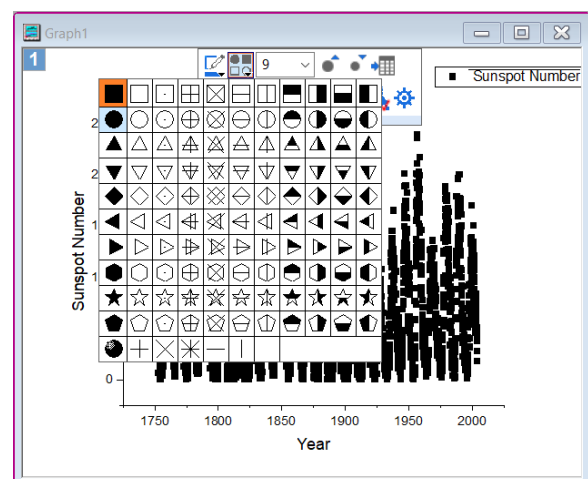
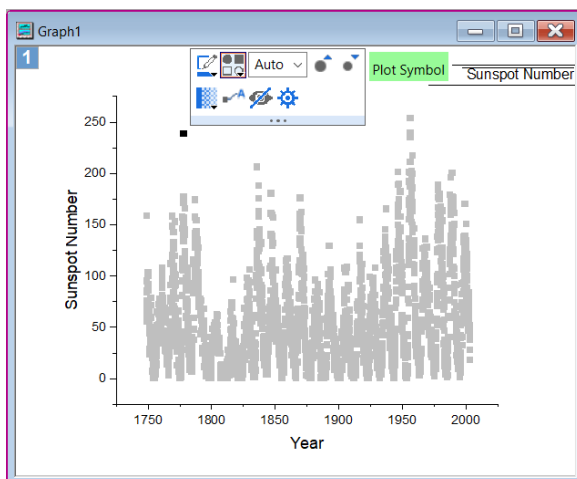
- II. To plot the sunspot number with respect to the year as a scatter plot, we click on the header of column C to select the entire column. Then from the main menu, we click Plot > Basic 2D > Scatter.



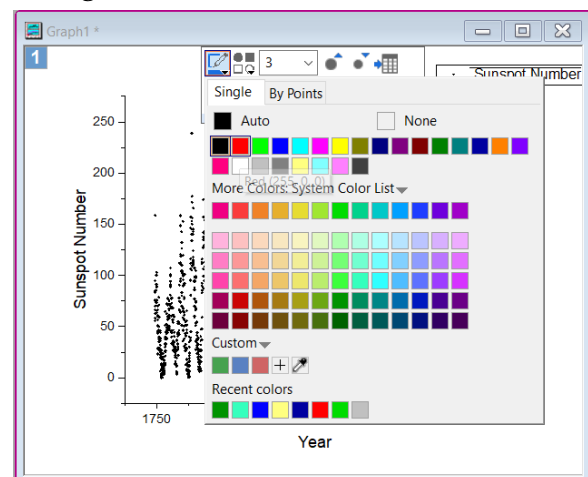
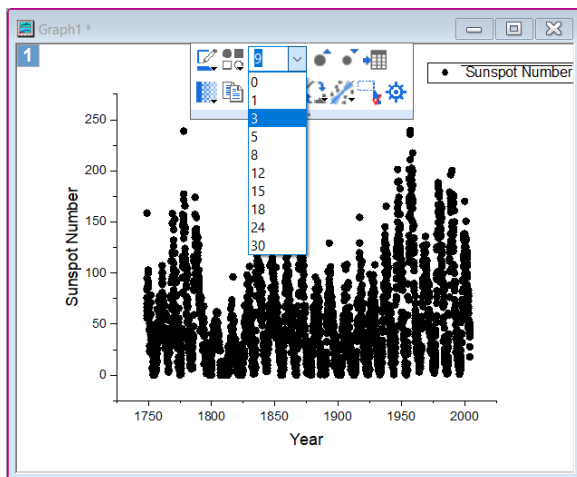
III. This gives us the default scatter plot with black squares at each data point.



IV. Its conventional to use coloured dots instead of black squares at each data point. To do so, we click on any one of the data points and from the pop-up menu, we click on the Plot Symbol button. Then we select the filled circle from the list of available symbols.



V. Next, we make the size of the circle smaller and change the colour to red.



VI. This finally gives us the scatter plot as expected.

